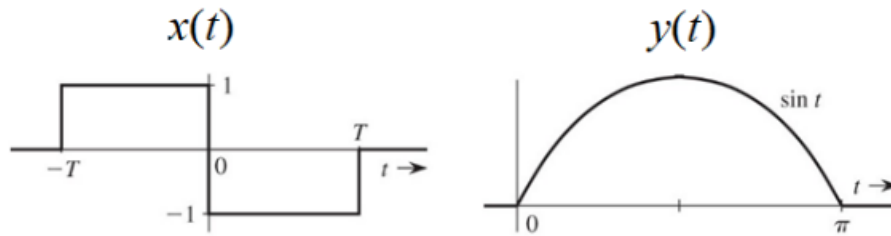


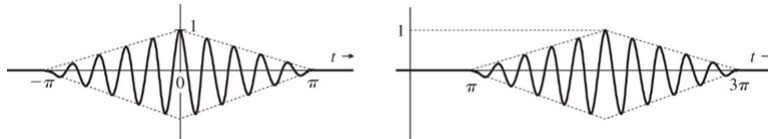
ENGR 3323: Signals and Systems

HW 11_Ch7

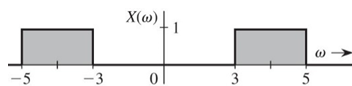
Q1) Find the Fourier transforms (FT) of the signals shown below using only the time-shifting property and the FT Table.



Q2) The signals below are modulated signals with carrier $\cos(10t)$. Find the Fourier transforms of these signals by using the Fourier transform properties and the FT table. Sketch the amplitude and phase spectra for the signals.



Q3) Use the frequency-shifting property and the FT Table to find the inverse Fourier transform of the spectra shown below



Q4) For a stable LTIC system with transfer function $H(s) = 1/(s + 1)$ find the (zero-state response) if the input $x(t)$ is

a) $e^{-2t}u(t)$

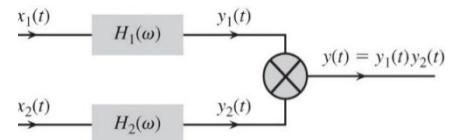
b) $e^{-t}u(t)$

c) $e^t u(-t)$

d) $u(t)$

Q5) Signals $x_1(t) = 10^4 \text{rect}(10^4 t)$ and $x_2(t) = \delta(t)$ are applied at the inputs of the ideal low-pass filters shown below $H_1(\omega) = \text{rect}(\omega/40,000\pi)$ and $H_2(\omega) = \text{rect}(\omega/20,000\pi)$. The outputs $y_1(t)$ and $y_2(t)$ of these filters are multiplied to obtain the signal $y(t) = y_1(t)y_2(t)$.

- a) Sketch $X_1(\omega)$ and $X_2(\omega)$. b) Sketch $H_1(\omega)$ and $H_2(\omega)$.
c) Sketch $Y_1(\omega)$ and $Y_2(\omega)$. d) Find the bandwidths of $y_1(t)$, $y_2(t)$, and $y(t)$.



Q6) For the signal $x(t) = \frac{2a}{t^2 + a^2}$ determine the essential bandwidth B (in hertz) of $x(t)$ such that the energy contained in the spectral components of $x(t)$ of frequencies below B Hz is 99% of the signal energy E_x .

Q7) Figure below shows a scheme to transmit two signals $m_1(t)$ and $m_2(t)$ simultaneously on the same channel (without causing spectral interference). Such a scheme, which transmits more than one signal, is known as signal *multiplexing*. In this case, we transmit multiple signals by sharing an available spectral band on the channel, and, hence, this is an example of the *frequency-division* multiplexing. The signal at point b is the multiplexed signal, which now modulates a carrier of frequency 20,000 rad/s. The modulated signal at point c is now transmitted over the channel.

- a) Sketch the spectra at points a , b , and c . b) What must be the minimum bandwidth of the channel?
c) Design a receiver to recover signals $m_1(t)$ and $m_2(t)$ from the modulated signal at point c .

